

Write Disturb in Ferroelectric FETs and Its Implication for 1T-FeFET AND Memory Arrays

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Abstract—In this letter, the write disturb of $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ -based 1T-FeFET nonvolatile AND memory array is experimentally investigated for $V_W/2$ and $V_W/3$ inhibition bias schemes to determine the worst-case memory sensing condition. Read margin analysis reveals that the increased leakage current in the low- V_{TH} erased state and the increased read current of the high- V_{TH} programmed state are the key factors that limit the maximum array size.

Index Terms—HZO, FeFET, memory array, write disturb.

I. INTRODUCTION

THE excellent scalability and CMOS compatibility of HfO_2 -based ferroelectrics elucidates their use in ferroelectric-field-effect-transistors (FeFET) for application in embedded non-volatile memory (NVM) [1], [2]. Ever since the discovery of ferroelectricity in HfO_2 , significant progress has been made in the design and fabrication of FeFET memory.

The major challenges for FeFET memory are the limited endurance of a single cell and write disturb in a 1T-FeFET array. Due to either charge trapping in pre-existing traps and generated defects or dielectric breakdown, HfO_2 FeFETs exhibit limited endurance cycles [6], [7]. Strategies to improve endurance have been proposed by increasing the interlayer thickness or distributing the electric field evenly between the ferroelectric and the semiconductor [8], [9].

While the 1T-FeFET memory array is attractive for embedded NVM application for write energy efficiency, high integration density (compared with other embedded NVMs), and non-destructive read, the absence of an additional selection transistor increases its susceptibility to write disturb for the half-selected memory cells in the array. To minimize the disturb effects, inhibition bias schemes such as $V_W/2$ and $V_W/3$ (V_W is write pulse amplitude), are usually applied across the half-selected cells during the memory write operation. This disturb effect has been studied with a high V_W of ± 5 V and large pulse width of 10 μs , which exacerbates the write disturb and even causes state collapse [10]. Therefore, it is imperative

to evaluate the write disturb for array-level operation for write pulses with smaller amplitude and shorter duration.

The write disturb effect on how the disturbed memory cells interfere with memory sensing is largely unexplored. It is known that write disturb shifts the device I_D - V_G characteristic, but the key parameters, the cell leakage current or read current, that deteriorate memory sensing need to be identified and is necessary for memory array optimization. Therefore, in this work we emulate the write disturb with ± 4 V 500ns write pulses for the two inhibition bias schemes in a single 1T-FeFET memory cell and study the impact of write disturb on an AND array architecture based on the single cell response.

II. 1T-FEFET ARRAY

There are a few FeFET memory array topologies [11]. The 1T-FeFET AND array can be more promising than other types of architectures, e.g. NOR array, for embedded NVM applications: (i) The cell size in an AND array is similar to or even smaller than that in a NOR array [12]; (ii) In the AND architecture in Fig. 1, the bit-line (BL) and the source-line (SL) are parallel, reducing the write disturb. In contrast, in a NOR array, the WL and SL run parallel, causing all the cells with the same WL (target cells and unselected cells) experience the same write voltage V_W , exacerbating the write disturb. Additionally, large channel current flows during memory write, increasing the write power consumption in a NOR array [11].

The two inhibition bias schemes, $V_W/2$ and $V_W/3$, for an AND memory array configuration, are illustrated in Fig. 1 (a). The reduction of disturb voltage amplitude in $V_W/3$ scheme comes at the cost of increasing the number of disturbed cells. For the $V_W/2$ bias scheme, the half-selected cells (which share either the same WL or the same BL/SL as the target cell) experience a disturb voltage of $V_W/2$ while the unselected cells experience no disturb. For $V_W/3$ bias scheme, in addition to the half-selected cells, which experience a disturb voltage of $V_W/3$, unselected cells experience disturb voltage of $-V_W/3$.

Disturbed memory cells will interfere with the memory read operation, potentially causing read errors. Fig. 1 (b) shows a typical row-wise current-mode sensing scheme for such an array. Like other current-mode sensing schemes [13], a reference sensing current I_{REF} between I_{ERS} and I_{PGM} is adopted to determine whether the target cell has a low- V_{TH} state or a high- V_{TH} state. The memory read access sensing margin analysis shows the practical I_{SENSE} range (Fig. 1 (c)). When accessing a high- V_{TH} state cell, the worst read case corresponds to the cell configuration where all the unselected cells are in the low- V_{TH} state, contributing additional leakage current, $(N-1) \cdot (I_{ERS})_L$, for a N -row 1T-FeFET array. As the

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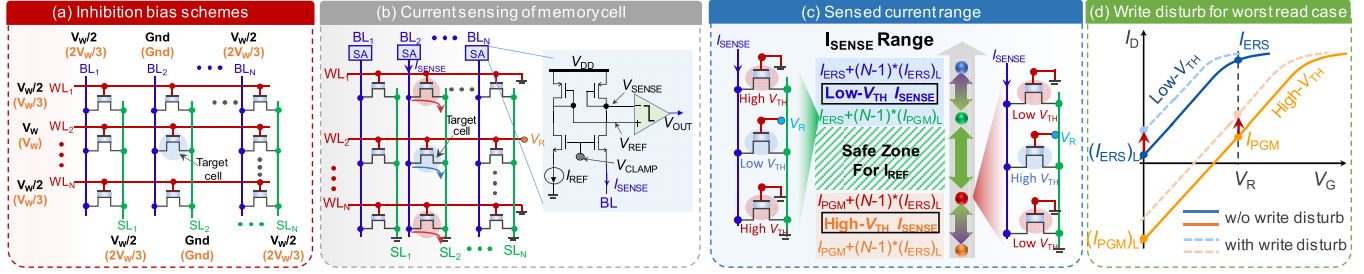


Fig. 1. (a) $V_W/2$ (black) and $V_W/3$ (yellow) write disturb inhibition bias schemes in a 1T-FeFET AND memory array. Here the target cell is shown in blue circle. For $V_W/2$ scheme, half-selected cells share either the same WL or the same BL as the target cell experience a disturb voltage of $V_W/2$. For $V_W/3$ scheme, in addition to half-selected cells which experience a disturb voltage of $V_W/3$, unselected cells experience a disturb voltage of $-V_W/3$. (b) Current-mode memory sensing in an 1T-FeFET memory array. Leakage current from unselected cells in the same column contributes to the total sensed current. (c) Sensed current I_{SENSE} range when reading a high V_{TH} or low V_{TH} device. The cell configuration for worst read situation is also shown. (d) Write disturb effect on FeFET characteristics for the worst read situation.

array size N increases, the leakage current may become larger than the target cell current, I_{PGM} , causing a reduction in sensing margin for a given I_{REF} . The write disturb conditions that increase the I_{PGM} or $(I_{ERS})_L$ are of the main interest, as shown in Fig. 1 (d), as they increase the total sensed current I_{SENSE} for high- V_{TH} state sensing and limit the maximum array size N .

When accessing a low- V_{TH} state memory cell, the write disturb is less of an issue, as the minimum sensed current corresponds to the cell configuration where all the unselected cells are in high- V_{TH} states. The total leakage current, $(N-1) \cdot (I_{PGM})_L$, add together with the target cell current I_{ERS} . Therefore, the increase of array size N is beneficial for low- V_{TH} state sensing. The degradation in low- V_{TH} state sensing comes mainly from the reduction of I_{ERS} , which arises only from disturb pulses and is not of concern, as shown in section III. Therefore, we mainly focus on the degradation of I_{PGM} and $(I_{ERS})_L$, which affects the high- V_{TH} state sensing.

Depending on the FeFET initial state, the disturb pulses with different polarities (positive or negative) will have different effects. Two competing mechanisms, polarization switching and charge trapping, collectively determine the disturb effects. For example, positive disturb pulses cause partial polarization switching for the high- V_{TH} state, reducing the device V_{TH} , while the induced electron trapping in the gate dielectric acts against the polarization and increases the V_{TH} [14], [15]. **Note that a ferroelectric film contains multiple domains with a distribution of coercive fields, and, hence, the disturb pulses with amplitude smaller than the cumulative coercive voltage can still induce partial polarization switching [16].** In the following, experimental results of write disturb effect on a single cell and its implication on the array size N will be analyzed.

III. RESULTS AND DISCUSSION

A. Write Disturb to Cell in Erased State (Low- V_{TH})

The device under test (DUT) features a gate-last process and a gate stack of $\text{TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{SiO}_2(0.8\text{ nm})/\text{Si}$. All the devices have a dimension of $W/L=20\mu\text{m}/1\mu\text{m}$ [15]. The write disturb to FeFET in the erased state and its effect on I_{ERS} and $(I_{ERS})_L$ are emulated by applying gate pulse sequence shown in Fig. 2. The delay T_{delay} between consecutive disturb pulses is set at 10 ms to minimize the effect of charge trapping on the device state [15]. Note that this emulation probably overestimates the write disturb effect since the memory cells will likely be disturbed by random bipolar pulses (positive or negative) in a real array, instead of identical pulses all the time.

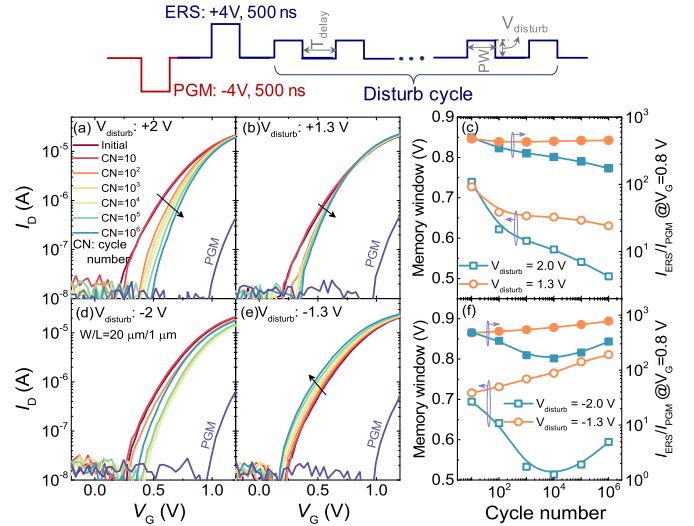


Fig. 2. Erased state I_D - V_G characteristic evolution with the increase of write disturb cycle number for disturb voltage, V_{disturb} of (a) +2 V; (b) +1.3 V; (d) -2 V; and (e) -1.3 V. (c)/(f) Memory window and I_{ERS}/I_{PGM} ratio evolution with the cycle number for positive/negative disturb pulse, respectively. All the applied pulses, including the program/erase pulse and the write disturb pulse, have a pulse width of 500 ns. Current measurement noise floor is at 10^{-8} A .

The effects of write disturb on the erased state I_D - V_G characteristic with disturb voltage of +2.0 V and +1.3 V are shown in Fig. 2 (a) and (b). Increasing V_{TH} with disturb cycle is indicative of electron trapping induced by the positive disturb pulses, as no polarization switching is happening for low- V_{TH} state under positive pulses. In this case, disturb pulses decrease $(I_{ERS})_L$, beneficial for high- V_{TH} state sensing. On the other hand, I_{ERS} is reduced by 65% after 10^6 cycles for +2.0 V disturb, and remains well above I_{REF} without causing sensing errors (I_{REF} is chosen to be the current at $V_G = V_R$ for a I_D - V_G curve with a V_{TH} equal to the average V_{TH} of programmed and erased state).

The disturb effect on the erased state with -2.0 V and -1.3 V pulses is shown in Fig. 2 (d-e-f). The decrease of V_{TH} with disturb cycle for -1.3 V results from hole trapping in the gate stack. This improves I_{ERS} , but also increases the leakage current $(I_{ERS})_L$, degrading high- V_{TH} state sensing. For disturb voltage of -2.0 V, the device V_{TH} increases initially, and then decreases beyond the 10^4 disturb cycle. This is caused by the competing effects between polarization switching and hole trapping. Initially, the polarization switching dominates, increasing the device V_{TH} , while with increasing disturb cycles, hole trapping in the pre-existing and newly generated

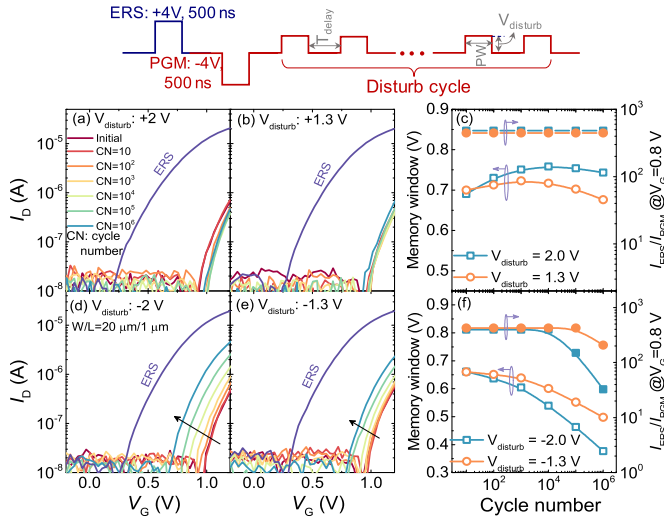


Fig. 3. Programmed state I_D - V_G characteristic evolution with the write disturb cycle number for disturb voltage, V_{disturb} of (a) +2 V; (b) +1.3 V; (d) -2 V; and (e) -1.3 V. (c)(f) Memory window and $I_{\text{ERS}}/I_{\text{PGM}}$ ratio evolution with the cycle number for positive and negative disturb pulse.

defects finally exceeds the polarization switching induced V_{TH} increase. The degradation in I_{ERS} (66% after 10^6 cycles) is small and does not cause any sensing error. In contrast to the continuous V_{TH} shift in the tested large devices, it has been shown that the disturb pulses can cause abrupt memory state transition for ultra-scaled devices (28nm FeFET) due to the limited number of domains [17]. Whether the abrupt switching is more detrimental than gradual switching needs further systematic evaluation since the scaled device V_{TH} remains unchanged until a threshold number of pulses are applied, suggesting that the array would not be degraded before that critical number. It's likely that operational and system optimization may benefit from this property.

B. Write Disturb to Cell in Programmed State (High- V_{TH})

By applying the gate pulse sequence, shown in Fig. 3, the write disturb to the DUT in the programmed state and its effect on I_{PGM} are studied. For positive disturb voltages, the device V_{TH} shifts positively in the beginning and then decreases beyond the 10^4 disturb cycles. This is the result of competition between electron trapping in the gate dielectric and polarization switching. The disturb effect in I_{PGM} is negligible and below the noise floor. Therefore, positive disturb pulse has a negligible effect on the sensing of high- V_{TH} state cell. The write disturb with negative pulses to the programmed state, however, is significant, as shown in Fig. 3 (d-e-f). The device V_{TH} shift negatively with the disturb cycle due to hole trapping, and thus I_{PGM} in the high- V_{TH} state increases significantly.

The polarization switching is observed at smaller disturb cycles for negative disturb pulses to low- V_{TH} state, than positive disturb pulses to high- V_{TH} state. This is likely related to the ferroelectric electric field asymmetry in the two cases. As the V_{TH} is positive for both low- V_{TH} and high- V_{TH} states, a larger electric field exists for the case of negative disturb pulse to low- V_{TH} state [15]. Considering the exponential dependence of polarization switching delay on the electric field [18], [19], it is easier to flip the polarization for negative pulses.

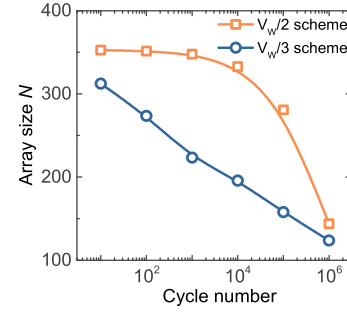


Fig. 4. Maximum array size N evolution with the disturb cycles for the worst read case of sensing a target cell in the high- V_{TH} state while all the other cells sharing the same bit line are in the low- V_{TH} state. For the analysis, the I_{REF} is chosen to be the current at $V_G = V_R$ for a $I_D - V_G$ with V_{TH} equal to the average V_{TH} of erased and programmed states. The maximum array size N is determined to satisfy $I_{\text{PGM}} + (N - 1) \times (I_{\text{ERS}})_L \leq I_{\text{REF}}$, which ensures no sensing errors.

C. Array Analysis

The response of a single memory cell to write disturb pulses illustrates two write disturb scenarios that would likely cause read errors when sensing a high- V_{TH} state cell for both $V_{\text{W}}/3$ and $V_{\text{W}}/2$ inhibition schemes. One is -1.3 V disturb to the low- V_{TH} state, which increases $(I_{\text{ERS}})_L$, and the other one is -2.0 V disturb to high- V_{TH} state, which increases I_{PGM} . Fig. 4 analyzes N as a function of the disturb cycle for the two cases. It is important to note that, unlike a passive crossbar array [20], the cumulative leakage currents of all unaccessed FeFETs can be further suppressed by applying a negative V_G , such that the array size N can be significantly increased. Without loss of generality, the gate electrodes of unaccessed cells are all grounded here.

An analysis of the impact of I_{PGM} increase on the array size N reveals that, for disturb cycle below 10^3 , N remains unchanged as the I_{PGM} is much smaller than I_{REF} . However, the array size N degrades significantly beyond 10^3 cycles (60% reduction at 10^6 cycles). The increase in $(I_{\text{ERS}})_L$ for -1.3 V disturb is monotonic with the disturb cycles. The array size reduces with cycling since the total leakage current, $(N - 1) \times (I_{\text{ERS}})_L$, is proportional to $(I_{\text{ERS}})_L$.

The increase in both I_{PGM} and $(I_{\text{ERS}})_L$ is caused by hole trapping induced by negative disturb pulses to both the programmed and erased state, which degrades the sensing of memory cell in the high- V_{TH} state in an 1T-FeFET array. Therefore, process optimization is necessary to reduce defect density and suppress hole trapping, in order to have a functional and robust 1T-FeFET array. In addition, circuit-level techniques, such as I_{REF} optimization and high-gain low-input-offset sensing schemes, would be useful for reducing the sensing error rate.

IV. CONCLUSION

We have demonstrated the write disturb effects in an 1T-FeFET array with $V_{\text{W}}/2$ and $V_{\text{W}}/3$ inhibition bias schemes. Charge trapping and polarization switching are two competing mechanisms that determine the impact of the disturbs. Worst-case read margin analysis reveals that leakage current increase of the low- V_{TH} erased state or the read current increase of the high- V_{TH} state current induced by disturb pulses are the limiting factors. This work elucidates the factors that require optimization and tuning for more robust read operations, to address the critical write disturb issue for the design of 1T-FeFET memory array.

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